Demo: Eddy Currents

ChatGPT edition — falling magnets, no electricity supply

Magnetic braking and energy accounting

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

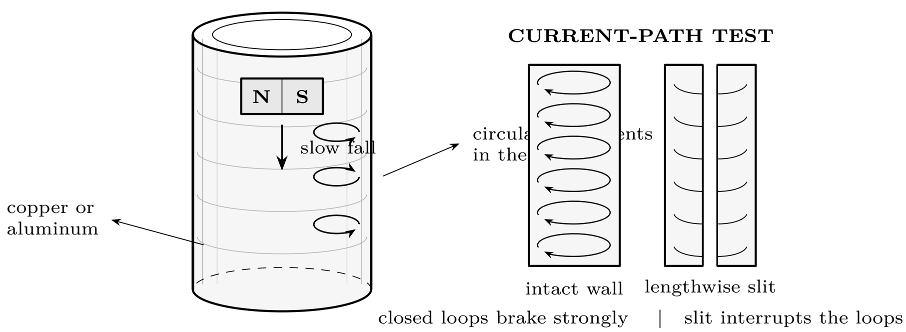
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Copper or aluminum tubes of known length; optional commercially prepared slit tube
- Same-bore nonconducting tube
- One captured or individually controlled neodymium magnet
- Same-size nonmagnetic slug
- Padded catch tray, ruler, timer, masking-tape labels, and room-temperature thermometer

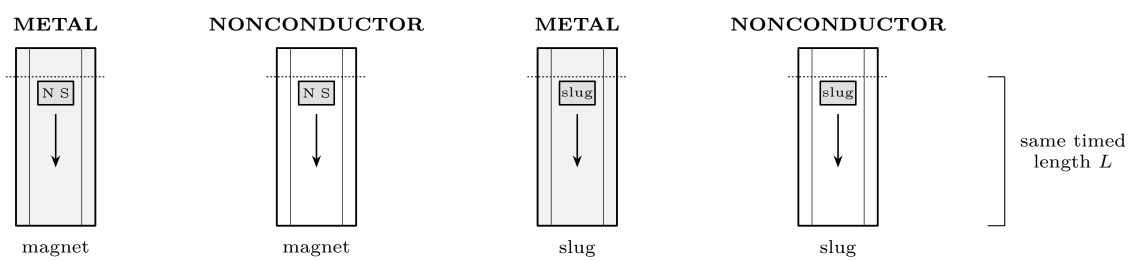
Predict first

Rank the transit times for the four pairings: magnet/metal, magnet/nonconductor, slug/metal, and slug/nonconductor. Which comparison isolates conductivity, and which isolates magnetism?



Investigate

1. Adult inspects the tubes, counts and controls the single magnet, and places the padded tray below the exit. Label tubes without hiding a slit.
2. Hold each tube vertical with its lower end at the same marked height. Release—do not push—the slug or magnet with its lower face level with the same start line. One person releases; another times first motion to first exit.
3. Run the four baseline pairings below in counterbalanced order. Record **five** trials per pairing. If a release or catch is disturbed, mark the trial invalid rather than silently replacing its value.



FACTORIAL CHECK Change one factor at a time; keep L, bore, release point, orientation, timer, and catch height fixed.

Baseline timing record · seconds

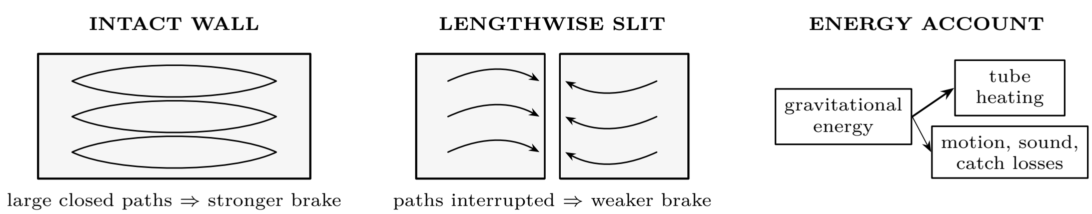
Pairing	1	2	3	4	5	Median	Notes / invalid trial
Magnet + metal							
Magnet + nonconductor							
Slug + metal							
Slug + nonconductor							

Reason before calculating. Which two rows test whether ordinary rubbing or air drag explains the slow fall? Which two rows show that a conductor alone is insufficient? Why is the median more resistant than the mean to one late button press?

4. Calculate each median by ordering the five valid times and selecting the middle value. Repeat a set if fewer than five trials remain valid.
5. If supplied, compare an intact conductor with a commercially prepared, burr-free lengthwise-slit conductor. Never cut or deburr tubing during this demo; the adult rejects sharp, crushed, or damaged specimens.

Explain from the evidence

Changing magnetic flux drives circulating currents in the tube wall. By Lenz's law their magnetic effect opposes the change that produced them, so the falling magnet experiences an upward braking force. Gravity still transfers energy: the magnet's lost gravitational potential energy becomes mainly thermal energy in the conductor, with smaller sound and motion losses. The metal need not be ferromagnetic. A nonmagnetic slug supplies the drag control; a nonconducting tube supplies the magnetic control. A lengthwise slit interrupts the broad closed current paths and therefore weakens the braking.



Claim check

If the magnet is slow because it "sticks to metal," what should happen with aluminum or copper? If energy disappears, what observation would violate energy conservation? State the evidence that distinguishes each mistaken claim from the eddy-current explanation.

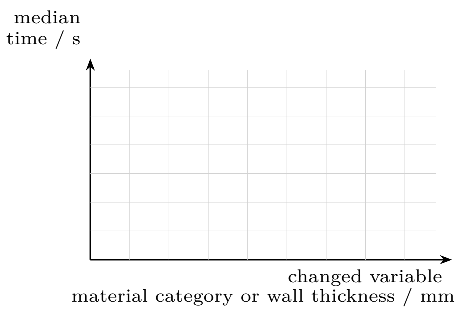
Change one thing

Material and wall test. Use only adult-supplied, undamaged tubes with the same timed length and closely matched bore. Vary one property per series. Record actual dimensions; do not infer thickness from apparent weight.

Controlled extension record

Tube ID	Material	Wall / mm	t_1	t_2	t_3	Median	Fixed / anomaly

Plot the evidence. Put the deliberately changed variable on the horizontal axis and median transit time on the vertical axis. Plot individual points; connect them only when the horizontal variable is ordered (for example, wall thickness). Material names are categories, so use separated points or bars.



Diagnose before repeating

Symptom	Check first
All times vary widely	release line, tube vertical, timer cue
Only magnet/metal varies	magnet turns or touches bore
Every pairing is slow	bore obstruction or catch cue
Slit tube brakes strongly	slit hidden, bridged, or mislabeled
Thicker tube is faster	length/bore changed too; tube warm

Final proof. Write one conclusion containing: the comparison that identifies both magnetism and conductivity as necessary; a numerical median with units; the direction (not merely the existence) of a material, thickness, or slit effect; one plausible uncertainty; and the energy destination. Do not substitute loose stacks of ingestible magnets.

Evidence record

Prediction ______________________________

One change ______________________________

Observation ______________________________

Claim + because ______________________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.